

B.Tech. Degree IV Semester Examination April 2019

ME 15-1403 MECHATRONICS (2015 Scheme)

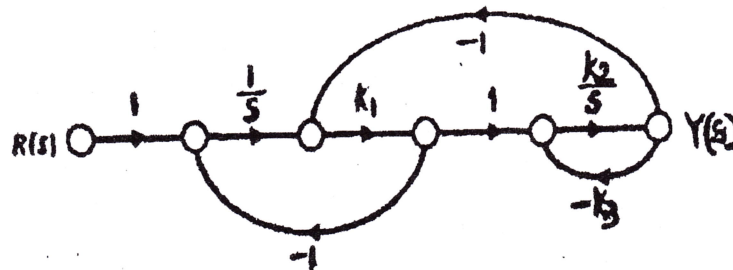
Time : 3 Hours

Maximum Marks : 60

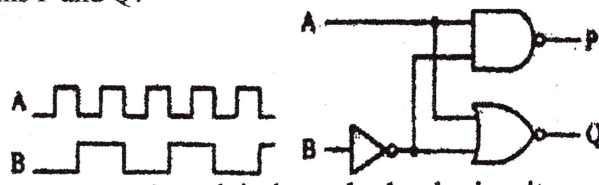
PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Define Mechatronics? Discuss on the scope of Mechatronics?
 (b) Explain the working of Thermistors and RTDs?
 (c) Explain Automatic Control System using a neat figure?
 (d) Describe the basic building blocks for the modeling of Thermal systems?
 (e) Describe a proportional plus derivative control? Mention its advantages.
 (f) Determine the closed loop transfer function $T(s) = (Y(s)/R(s))$ for the system shown in figure.



- (g) By routh stability criterion determine the stability of the system represented by the characteristic equation, $9s^5 - 20s^4 + 10s^3 - s^2 - 9s - 10 = 0$. Comment on the location of roots of the characteristic equation.
 (h) Explain full-stepping and half-stepping in a stepper motor?
 (i) Differentiate a microprocessor and microcontroller?
 (j) The figure shows the input signals A and B and the gate system to which it is applied. Draw the output waveforms P and Q?



PART B

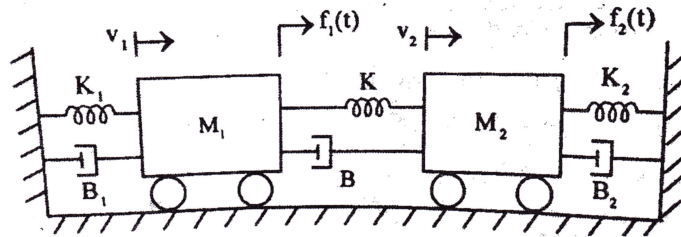
(4 × 10 = 40)

- II. Briefly explain Electric Transducers? What are the different classifications of Electric Transducers? Explain Resistive type Transducers and its applications using a neat figure (10)

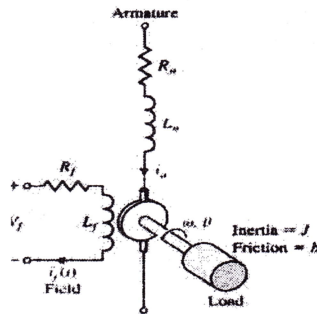
OR

(P.T.O.)

- III. (a) Explain Servo Mechanism? Explain the role of Servomotors in Mechatronics? (3)
 (b) Explain (i) Piezoelectric sensor (ii) LVDT (4)
 (c) Describe how a spool valve controls the direction of flow in hydraulic circuits? (3)
- IV. (a) Determine the force-voltage analogous circuit for the system shown in the figure. (5)



- (b) Explain PID controller and its advantages. Derive the transfer function of op-amp PID controller. (5)
- OR**
- V. (a) Obtain the transfer function of armature controlled d.c. motor whose wiring diagram is shown in the figure. (5)



- (b) Obtain the response of unity feedback system whose open loop transfer function is $G(s) = \frac{4}{s(s+5)}$ and when the input is unit step. (5)
- VI. Plot the Bode diagram for the following transfer function and obtain the gain and phase cross-over frequencies. (10)

$$G(s) = \frac{10}{s(1+0.4s)(1+0.1s)}$$

- OR**
- VII. A unity feedback control system has an open loop transfer function, $G(s) = \frac{K}{s(s^2+4s+13)}$. Sketch the root locus. (10)

- VIII. (a) Explain how an electronically controlled automobile engine is different from a conventional engine. What are its advantages? (5)
 (b) What is a Flip-Flop? Explain how the S-R flip-flop functions. Mention one use of flip-flop. (5)

OR

- IX. (a) A water pump for filling a tank is started using a spring-return push-button switch. When a sensor detects that the tank is full, the pump is stopped. Then the water is heated for 300 seconds using a heater. Draw the ladder diagram to represent this. (5)
 (b) Draw and explain the design a robotic arm for pick and place application, where each joint is actuated by separate pneumatic cylinders. (5)
